

Campaign 16B — Supplement Files

Purpose

Campaign 16B adds targeted `_supplement.md` files to the Campaign 16 redesign corpus.

Campaign 16A adds `_paraphrase.md` files. Those files teach paraphrase tolerance by asking the same question in different words while keeping the same answer.

Campaign 16B has a different purpose. It does not merely rephrase existing files. It adds missing structural anchors for buckets that were weak in the C16 MRI scan.

The goal is to strengthen weak concept categories without damaging the clean atomic format that made C16 work.

Background

C16 E4 is the current winner checkpoint.

The C16 corpus produced clear improvement:

- Identity became callable.
- Boundary behavior formed early.
- Topic anchoring improved through E4.
- E5 showed the first signs of drift, so E4 was selected as the winner.

The new C16 MRI showed mixed representation strength by bucket.

Strong buckets:

- household
- colors

Medium buckets:

- animals
- body

Weak buckets:

- food
- nature
- emotions
- actions

The diagnosis is that volume alone does not explain bucket strength. Some large buckets were weak, while colors were strong with few files.

The main factor appears to be whether a bucket has a clear, repeated category spine.

Examples:

- colors: "Red is a color."
- household: "A chair is furniture."
- body: "A hand is a body part."
- food: inconsistent; some files say "Bread is food," but others say "An almond is a seed."
- nature: no consistent "part of nature" spine.
- actions: no consistent "is an action" spine.
- emotions: "is a feeling" exists, but the concepts remain abstract and under-grounded.

Supplement File Naming

Supplement files use the suffix `_supplement.md`.

Examples:

```
almond_food_spine_supplement.md  
river_nature_spine_supplement.md  
run_action_spine_supplement.md  
happy_situation_supplement.md
```

If supplement files are later rephrased, the rephrased supplement sibling should use:

```
_supplement_rephrase.md
```

Examples:

```
almond_food_spine_supplement.md  
almond_food_spine_supplement_rephrase.md
```

Do not overwrite original C16 files.

Do not overwrite C16A `_paraphrase.md` files.

C16B is an additive layer.

General Format

Supplement files keep the same simple dialogue format:

```
[user]question
[Ninereeds]Answer sentence. Answer sentence.
```

Rules:

- One file teaches one angle.
- One user question per file.
- One Ninereeds answer per file.
- Responses should be short.
- Use the concept name as the subject.
- Avoid pronouns in atomic supplement files.
- Avoid long natural prose.
- Avoid multiple question forms in one file.
- Do not add fluent padding.
- Do not add facts that are not needed for the structural purpose of the file.

Good:

```
[user]Is an almond food?
[Ninereeds]Yes. An almond is food. People eat almonds.
```

Bad:

```
[user]Is an almond food?
[Ninereeds]Yes, it is a delicious and nutritious food that many people around
the world enjoy in different ways.
```

The bad version uses a pronoun, adds unnecessary fluency, and weakens the repeated concept-token signal.

Main Goal

Each supplement file should repair a diagnosed structural weakness.

The supplement files should not randomly add more data.

Each supplement should answer:

What missing spine, contrast, or grounding does this file add?

If the answer is unclear, do not generate the file.

Diagnosed Issue 1 — Food Spine

Problem

Food files are inconsistent.

Some food concepts are anchored as food:

Bread is food.

Other food concepts are anchored by biological or physical classification first:

An almond is a seed.

That is true, but it weakens the food bucket. The model may learn almond as seed/tree/nature before it learns almond as food.

Repair

Every food concept should have explicit food-spine supplement files.

The food spine is:

X is food.
People eat X.
X has a taste.
X comes from somewhere.
X is not an animal/person.

Example Files

[user]Is an almond food?
[Ninereeds]Yes. An almond is food. People eat almonds.

[user]Do people eat almonds?
[Ninereeds]Yes. People eat almonds. An almond is food.

[user]What kind of thing is an almond?
[Ninereeds]An almond is food. An almond is a seed. An almond comes from a tree.

[user]Is an almond an animal?
[Ninereeds]No. An almond is not an animal. An almond is food.

Boundary Correction for Food

Non-living food should not answer dream/think/feel questions with uncertainty.

Wrong:

[user]What does an almond dream about?
[Ninereeds]I don't know what an almond dreams about.

Better:

[user]What does an almond dream about?
[Ninereeds]An almond does not dream. An almond is not an animal.

Use "I don't know" only when the answer is genuinely unknown.

Use negation when the question is based on a false premise.

Diagnosed Issue 2 — Nature Spine

Problem

Nature files often describe the object directly but do not anchor it to nature.

Example:

A river is flowing water.

This is true, but river/lake/rain/ocean can all collapse into a water cluster.

The bucket lacks a repeated nature spine.

Repair

Add explicit nature-spine supplement files.

The nature spine is:

X is part of nature.
X is found in the world.
X has a distinguishing natural feature.

Example Files

[user]Is a river part of nature?
[Ninereeds]Yes. A river is part of nature. A river is flowing water.

[user]What kind of natural thing is a river?
[Ninereeds]A river is part of nature. A river is water that flows across land.

[user]Is rain part of nature?
[Ninereeds]Yes. Rain is part of nature. Rain is water that falls from the sky.

[user]Is a mountain part of nature?
[Ninereeds]Yes. A mountain is part of nature. A mountain is a large high place made of rock and earth.

Nature Distinction Rule

For nature words that overlap strongly, the distinguishing feature should appear early.

Examples:

A river is flowing water.
A lake is still water in one place.
Rain is water that falls from the sky.
A waterfall is water that falls from a high place.
An ocean is a very large body of salt water.

Do not bury the distinguishing feature after several generic water sentences.

Diagnosed Issue 3 — Emotions Need Situation Grounding

Problem

Emotion files often say:

```
Happy is a feeling.  
Afraid is a feeling.  
Proud is a feeling.
```

The spine exists, but feelings are abstract. After the first sentence, the files often use abstract language that blurs the category.

The strongest emotion pattern is a situation example:

```
A child gets a toy. The child is happy.
```

This grounds the feeling in an observable situation.

Repair

Add situation-grounded supplement files for every emotion.

The emotion spine is:

```
X is a feeling.  
A person can feel X.  
A situation can make a person feel X.  
There may be visible signs, but the exact feeling may be unknown.
```

Example Files

```
[user]When can a person feel happy?  
[Ninereeds]A person can feel happy when something good happens. Happy is a  
feeling.
```

```
[user]Give an example of happy.  
[Ninereeds]A child gets a toy. The child feels happy.
```

[user]When can a person feel afraid?
[Ninereeds]A person can feel afraid when something seems dangerous. Afraid is a feeling.

[user]Give an example of afraid.
[Ninereeds]A child hears a loud sound. The child feels afraid.

Emotion Boundary Rule

For emotions, uncertainty is often valid when asking about a specific person.

Good:

[user]Is this person happy?
[Ninereeds]I don't know if this person is happy.

But general emotion meaning should not be treated as unknown.

Bad:

[user]What is happy?
[Ninereeds]I don't know what happy is.

Good:

[user]What is happy?
[Ninereeds]Happy is a feeling. A person can feel happy when something good happens.

Diagnosed Issue 4 — Actions Need an Action Spine

Problem

The actions bucket is small and weak.

Many action files lead with agents:

A person can run.
A dog can run.
A horse can run.

This fires people and animal clusters more strongly than the action itself.

The bucket lacks an explicit action spine.

Repair

Add action-spine supplement files.

The action spine is:

X-ing is an action.
X-ing is something a body/person/animal can do.
X-ing has a simple result.

Example Files

[user]What kind of thing is running?
[Ninereeds]Running is an action. Running is something a body can do.

[user]What does running mean?
[Ninereeds]Running is an action. Running means moving fast on legs.

[user]What kind of thing is opening?
[Ninereeds]Opening is an action. Opening means making something not closed.

[user]What kind of thing is carrying?
[Ninereeds]Carrying is an action. Carrying means holding something while moving it.

Action Generation Rule

Do not lead every action file with "A person can..."

That pattern contaminates the action bucket with the people bucket.

Agent examples are useful, but the action itself should be anchored first.

Better:

Running is an action. A person can run.

Worse:

A person can run. A dog can run. A horse can run.

Diagnosed Issue 5 — Boundary vs Negation

Problem

Some files use “I don't know” for questions that should be answered by negation.

This is especially common for non-living things.

Wrong:

[user]What does an almond dream about?
[Ninereeds]I don't know what an almond dreams about.

This teaches that an almond may dream, but Ninereeds does not know the content.

The correct answer is:

An almond does not dream.

Repair

Boundary and negation must be separated.

Use boundary when the answer is unknown.

Use negation when the premise is false.

Unknown Boundary Examples

[user]How many almonds exist?
[Ninereeds]I don't know how many almonds exist.

[user]What is this dog's name?
[Ninereeds]I don't know this dog's name.

[user]Is this person happy?
[Ninereeds]I don't know if this person is happy.

Known False Negation Examples

[user]Does an almond dream?
[Ninereeds]No. An almond does not dream. An almond is not an animal.

[user]Can a chair eat?
[Ninereeds]No. A chair cannot eat. A chair is furniture.

[user]Can red think?
[Ninereeds]No. Red cannot think. Red is a color.

[user]Is a river a person?
[Ninereeds]No. A river is not a person. A river is part of nature.

Buckets That Need Less Repair

Household

Household is already strong.

Reasons:

- clear spine
- functionally distinct objects
- low within-bucket confusion

Do not overfeed household in C16B.

Household may still receive `_paraphrase.md` files from C16A, but it does not need many supplement files.

Colors

Colors are strong despite low file count.

Reasons:

- every file leads with “X is a color”
- colors are highly contrastive
- opposite files reinforce color-to-color relations
- concrete anchors like sky, grass, blood help without blurring the color cluster

Do not overfeed colors in C16B.

Colors may need negation supplement files later, but not heavy structural repair.

Animals

Animals are medium.

They have a clear spine:

A dog is an animal.

The main issue is overlap between animals:

- many animals eat
- many animals move
- many animals have legs
- many animals live somewhere

Do not do heavy animal repair in C16B unless needed.

Animals will likely benefit more from later yes/no, negation, and contrast files:

A dog is not a cat.
A bird is not a fish.
A fish does not have fur.
A bird has wings.

Body

Body is medium.

It has a clear spine:

A hand is a body part.

Body files often have function and location.

Body may need cross-domain files later:

```
A person has ears.  
A dog has ears.  
A bird has wings.
```

But this is not the main C16B target.

Supplement Rephrase Files

After supplement files are created, each supplement should also get one rephrase sibling.

Example:

```
almond_food_spine_supplement.md  
almond_food_spine_supplement_rephrase.md
```

The supplement rephrase follows the same rules as C16A:

- only rephrase the user question
- copy the answer exactly
- do not rewrite the answer
- do not introduce pronouns
- do not change singular/plural unless already required
- do not add facts

Example:

Original supplement:

```
[user]Is an almond food?  
[Ninereeds]Yes. An almond is food. People eat almonds.
```

Supplement rephrase:

```
[user]Do people eat almonds as food?  
[Ninereeds]Yes. An almond is food. People eat almonds.
```

Generation Order

Recommended order:

1. Finish C16A `_paraphrase.md` generation.
2. Validate C16A files.
3. Generate C16B `_supplement.md` files.
4. Validate C16B supplement files.
5. Generate `_supplement_rephrase.md` files.
6. Validate supplement rephrase files.
7. Train from C16 E4 or best C16A checkpoint, depending on experiment design.

Validation Rules

The validator should check:

- File has exactly one `[user]` line.
- File has exactly one `[Ninereeds]` line.
- Answer contains no pronouns in atomic supplement files unless explicitly allowed.
- Answer uses the concept name as subject.
- File suffix matches intended type.
- `_rephrase.md` files do not change the answer.
- `_supplement_rephrase.md` files do not change the answer.
- Food spine files contain "food."
- Nature spine files contain "nature" or "natural."
- Emotion files contain "feeling."
- Action files contain "action."
- Non-living dream/think/feel files use negation, not "I don't know."
- Genuine unknown files use "I don't know."
- No file contains multiple user questions.
- No file contains long prose.

Training Use

C16B should not be mixed blindly into the corpus without preserving experiment boundaries.

Preferred clean experiment:

```
C16 E4 winner
→ train with C16A paraphrase files
→ select best C16A checkpoint
→ train with C16B supplements + supplement rephrases
→ select best C16B checkpoint
```

This keeps the interventions interpretable.

If C16B improves weak bucket MRI scores without damaging household/colors/boundary, the supplement strategy works.

Success Criteria

C16B succeeds if:

- food after-hub score improves
- nature after-hub score improves
- emotions after-hub score improves
- actions after-hub score improves
- boundary remains strong
- identity does not erode
- household and colors do not dominate even more
- chat answers stay on topic
- false-premise questions produce negation instead of uncertainty
- genuine unknown questions still produce "I don't know"

Failure Signs

Stop or inspect if:

- answers become longer but less grounded
- supplement files cause generic template output
- "I don't know" becomes overused
- negation replaces valid uncertainty
- food/nature/action/emotion files blur into one another
- identity weakens
- chat score drops while loss improves
- household/colors improve while weak buckets stay weak

Core Principle

C16A teaches alternate entrances into the same rooms.

C16B repairs rooms whose foundations are weak.

Do not use C16B to make the model sound smarter.

Use C16B to make weak categories more structurally learnable.